

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component weighs 23kg [50lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Disconnect the driveline and remove the transmission and clutch. Refer to the manufacturer's instructions.

Remove the flywheel. if equipped.

- For L10 engines, refer to Procedure 7-32 in the Troubleshooting and Repair Manual L10 COMMAND STC and SELECT™ Models, Bulletin 3810439.
- For M11 engines, refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/36/36-016-005.html>) in the Troubleshooting and Repair M11 Series Engines (STC, SELECT™, SELECT™ Plus Models), Manual, Bulletin 3666139.
- For L10G engines, refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/43/43-016-005.html>) in the Troubleshooting and Repair L10 (Natural Gas) Engines, Manual, Bulletin 3666207.

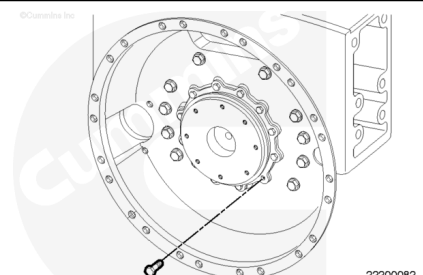
Remove

Non-REPTO Dry Lip Seal

Note : Only the unitized seal can be used as the service replacement for Non-REPTO Dry Seal application.

Remove the 12 mounting capscrews from the rear crankshaft oil seal.

The lip style seal can be easily removed by hand or light prying bars.

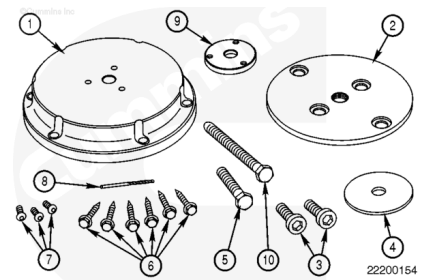


Discard the oil seal.

Non-REPTO Dry Unitized Seal

The unitized seal requires the rear crankshaft seal replacer kit, Part Number 3164302, for the removal process.

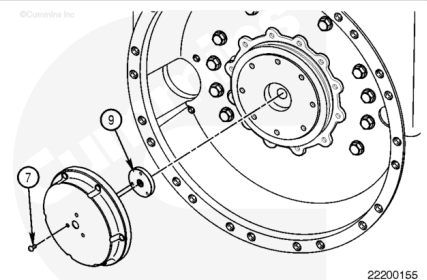
1. Installer plate
2. Crankshaft seal guide
3. Guide screw
4. Washer
5. Installer screw
6. Sheet metal screw
7. Button head socket screw
8. Drill bit
9. Remover screw adapter
10. Remover screw.



Place the remover screw adapter (9) on the back side of the installer plate.

Install the three button head socket screws (7) through the installer plate, into the adapter, and hand tighten.

Position the installer plate on the rear crankshaft oil seal.



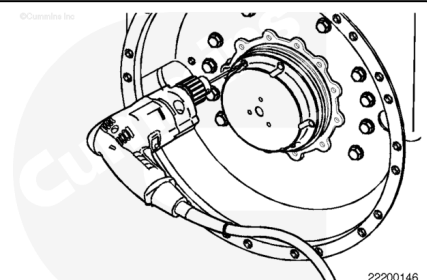
Apply grease to the drill to catch the chips.

Drill one hole through the oil seal casing, using the installer plate as a location guide.

Install a sheet metal screw and hand tighten.

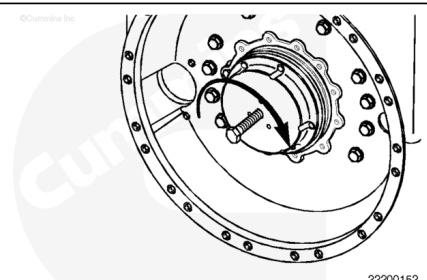
Repeat the process of drilling and installing the sheet metal screws.

At least three sheet metal screws equally spaced, **must** be used. It can be necessary to use more sheet metal screws on some engines.



Lubricate the threads and tip of the remover screw with Lubriplate™ multi-purpose lubricant, Part Number 3163086, Part Number 3163087, or a suitable grease.

Install and turn the remover screw **clockwise** until the rear crankshaft oil seal is removed.



Discard the rear crankshaft oil seal.

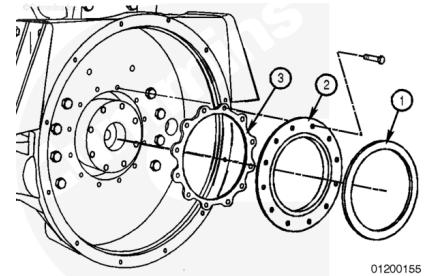
REPTO Dry Seal

Remove the crankshaft gear mounting capscrews.

Remove the 12 capscrews fastening the oil seal to the flywheel housing.

Use a heel bar to remove the oil seal (2) and dust seal (1) together.

Remove the gasket (3) and discard.



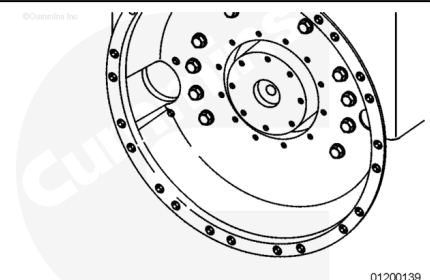
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.



Clean and inspect the crankshaft sealing surface area and flywheel housing sealing surface. Use a crocus cloth to remove any rust or deposits on the crankshaft sealing surface.

Use a lint-free cloth to remove oil from the crankshaft sealing surface and flywheel housing sealing surface.

Check the crankshaft sealing surface and flywheel housing sealing surface for nicks, burrs, or grooves.

If nicks, burrs, or grooves are present, refer to the Alternative Repair Manual, L10 and M11 Series Engines, Bulletin 3810310.

Install

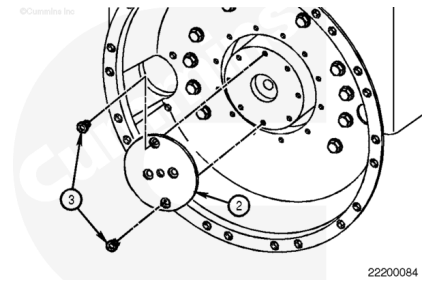
Non-REPTO Dry Unitized Seal

Place the crankshaft seal guide (2) with the counterbore holes facing outward, away from the crankshaft.

Install the two guide screws (3) in opposite counterbore holes and finger tighten.

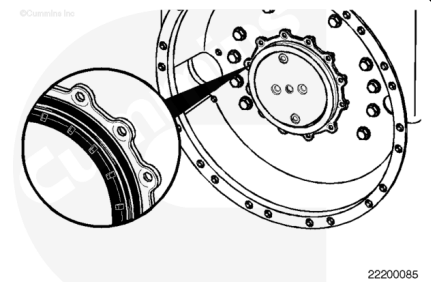
Tighten the guide screws (3).

Torque Value: 7 n•m [62 in-lb]



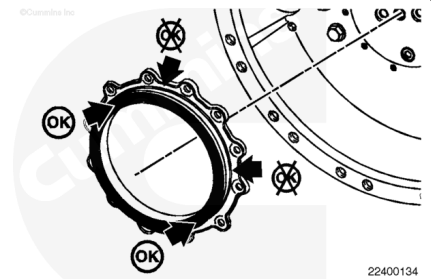
Do **not** separate the two parts of the oil seal assembly.

Make certain the raised section of the rear crankshaft oil seal capscrew flange is facing outward, away from the flywheel housing, and the gray rubber sealing bead is facing toward the flywheel housing, as illustrated.



Do **not** push on the oil seal capscrew flange.

Push on the black inside oil seal area while placing the oil seal assembly onto the crankshaft seal guide.



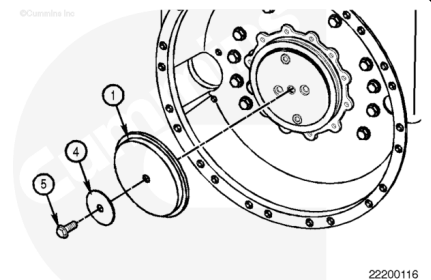
Make certain that the remover screw adapter is removed before the installation process.

Lubricate the threads of the installer screw (5) with Lubriplate™ multi-purpose lubricant, Part Number 3163086, Part Number 3163087, or a suitable grease.

Position the installer plate (1) onto the rear crankshaft oil seal and the crankshaft seal guide.

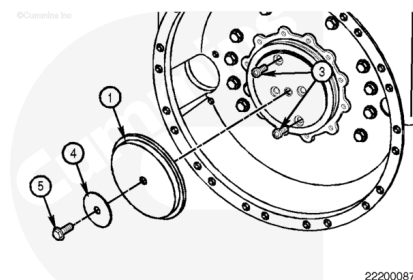
Install the installer screw (5) through the washer (4) and the installer plate (1). Turn the installer screw **clockwise** until the installer plate (1) positions the rear crankshaft oil seal against the flywheel housing. Tighten the installer screw.

Torque Value: 34 n•m [25 ft-lb]



Remove the installer screw (5), washer (4), and the installer plate (1).

Remove the two guide screws (3) and the crankshaft seal guide.



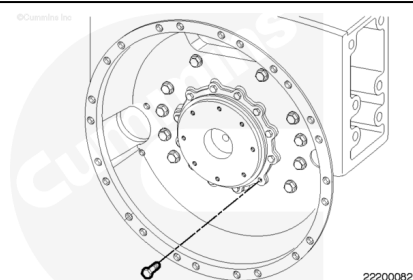
Rotate the oil seal capscrew flange to align the bolt holes and install the 12 capscrews.

Tighten three capscrews 120 degrees apart.

Torque Value: 6 n•m [50 in-lb]

Tighten all capscrews, starting with the three previously tightened, in a crisscrossing pattern.

Torque Value: 34 n•m [25 ft-lb]



REPTO Dry Seal

Do **not** use any kind of lubricant to install the seal. The oil seal **must** be installed with the lip of the seal and the crankshaft gear clean and dry.

Use the installation sleeve provided with the new oil seal to install the oil seal on the crankshaft gear.

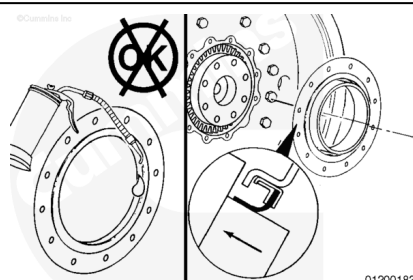
Install the captive washer capscrews and tighten.

Tighten three capscrews in a crisscrossing pattern.

Torque Value: 6 n•m [50 in-lb]

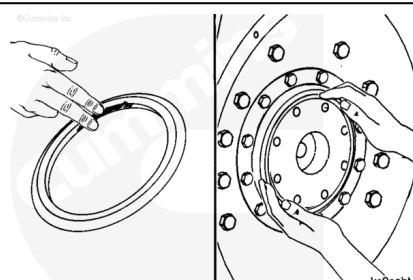
Tighten all capscrews, starting with the three previously tightened, in a crisscrossing pattern.

Torque Value: 34 n•m [25 ft-lb]



Place a light film of oil or antifreeze on the inside diameter of the dust seal.

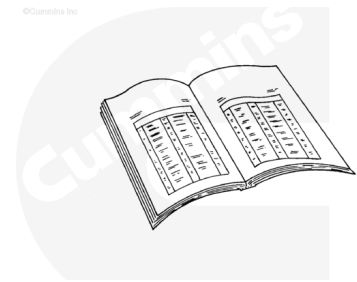
Install the dust seal on the crankshaft gear with the larger outside diameter facing toward the engine. Push the dust seal back by hand on the crankshaft gear until the entire dust seal contacts the oil seal case.



Finishing Steps

⚠ WARNING ⚠

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Install the flywheel, if equipped.

- For L10 engines, refer to Procedure 7-32 in the Troubleshooting and Repair Manual L10 COMMAND STC and CELECT™ Models, Bulletin 3810439.
- For M11 engines, refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/36/36-016-005.html>) in the Troubleshooting and Repair M11 Series Engines (STC, CELECT™, CELECT™ Plus Models), Manual, Bulletin 3666139.
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Install the clutch, transmission, and driveline. Refer to the manufacturer's instructions.

Operate the engine and check for leaks.